Brake Mode Experiment

What do our drive motors **really** do when the driver lets go of the sticks? (GitHub issue #17)

Team members		Date	
Robot		Battery voltage	

Equipment: robot + charged battery, Driver Station, tape measure, painter's tape, phone for photos/video. **Safety:** clear space at least 10 ft long, people stand to the side, a driver ready to press STOP.

1. What is brake mode?

Brake mode (in code: *ZeroPowerBehavior*) decides what a motor does when its power is **0**, like when the driver lets go of the sticks. It is a different setting from encoder mode (#11).

Mode	What the hub does	What you feel
BRAKE	Connects the motor's wires together. A spinning motor acts like a generator and fights its own spinning.	The robot stops quickly. Hard to push by hand.
FLOAT	Disconnects the motor. It spins freely.	The robot coasts to a stop. Easy to push.

BRAKE is **not** "hold position." It uses no power and only resists while the wheels are turning.

2. Our hypothesis (what we think happens)

Last season we assumed the motors BRAKE by default. But reading the FTC SDK's own code suggests:

a. When the robot **powers up**, the SDK sets all motors to **FLOAT**.

b. When a program (OpMode) **starts**, brake mode is **not** reset. The last program's setting **carries over**.

c. So **manualDriver** (which never sets brake mode) would **coast** after power-up or after Pinpoint Test, and only brake after a program that sets BRAKE.

Reading code is not proof. **We test it on the real robot.**

3. Predict FIRST (before you test!)

Circle your prediction for each step. Scientists write predictions down **before** the experiment, so the results can't change what we "remember" predicting.

Step	Prediction (circle one)
A1: brake mode left over after power-up	BRAKE / FLOAT
A2: brake mode left over after Pinpoint Test	BRAKE / FLOAT
A3: brake mode left over after Example TeleOp	BRAKE / FLOAT
B1: manualDriver after power-up	STOPS / COASTS
B2: manualDriver after Pinpoint Test	STOPS / COASTS
B3: manualDriver after Example TeleOp	STOPS / COASTS

Part A: What brake mode is left over?

Do the step, then run **Brake Mode Check** (Tests group) and press **INIT** only. Write down "**Brake mode when this OpMode started**". Then press START and STOP.

Note: Brake Mode Check itself leaves the motors in BRAKE when it stops. That's why each step starts with its own setup.

Step	Do this first...	Prediction (from page 1)	Result: BRAKE or FLOAT
A1	Power the robot OFF, then ON		
A2	Run Pinpoint Test: INIT, START, then STOP		
A3	Run Example TeleOp: INIT, START, then STOP		
A4	Run Pinpoint Test again: INIT, START, then STOP		

Part B: Does it change how manualDriver drives?

Do the setup, then run **manualDriver**. Drive forward at **full speed**, **let go** at a tape line, and measure how far past the line the robot rolls. Then, with the robot stopped, try pushing it.

Step	Setup before manualDriver	Prediction: stops / coasts	Roll distance (in)	Easy or hard to push?
B1	Power OFF, then ON			
B2	Run Pinpoint Test, then STOP			
B3	Run Example TeleOp, then STOP			

What do you notice? Does the same program (manualDriver) behave differently depending on what ran before it?

Part C: Feel the difference

Run **Brake Mode Check** and press START. Press **X** for BRAKE or **Y** for FLOAT. Mark a start line with tape. Drive forward at **full speed**, **let go** at the line, and measure how far past it the robot rolls. Do 3 tries in each mode.

Mode	Try 1 (in)	Try 2 (in)	Try 3 (in)	Average (in)
BRAKE (X)				
FLOAT (Y)				

Average = (Try 1 + Try 2 + Try 3) / 3

Question	BRAKE	FLOAT
With the robot stopped, how hard is it to push? (1 = easy, 5 = very hard)		
Driver 1: which one do you like driving? Why?		
Driver 2: which one do you like driving? Why?		

Think about the game: the field gets crowded. Which mode helps a driver stop exactly where they want (lining up to launch, parking in the LOADING ZONE)? Which mode helps when another robot pushes ours?

Results and decision

Was our hypothesis right? (circle) YES / NO / PARTLY

Explain what you found, using your numbers from Parts A, B and C:

The fix (a coder does this with a mentor)

In **PinpointTest.java**, at the end, un-comment this line so Pinpoint Test puts the motors back to BRAKE:

```
TestDriveMotors.setBrake(hardwareMap, true);
```

Rebuild, then repeat step B2 to show the problem is gone:

Step	Setup	Roll distance (in)	Stops or coasts?
B2 again	Run Pinpoint Test (with the fix), then STOP, then manualDriver		

Team decision

What should we do about programs that never set brake mode (like manualDriver)? Check one:

	Option
	manualDriver also sets BRAKE when it starts
	Drivers stop using manualDriver and use **Main TeleOp** (it already sets BRAKE at INIT)
	Something else: _____

Why?

After the test: a mentor records the decision in `TeamCode/docs/decisions.md`, the results go in GitHub issue #17, and this sheet (plus photos and the stopping video) goes into the portfolio entry `docs/portfolio/2026-09-26-brake-mode-experiment.md`. Great judging material: hypothesis, prediction, experiment, result, fix!